The coordinate ring of the k-fold iterated commutator locus for 2×2 matrices

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Abstract

For 2×2 matrices $A_1, ..., A_k$, write $[A_1, ..., A_k]$ for the left-normed iterated commutator $[...[[A_1, A_2], A_3], ..., A_k]$, and I_k for the ideal, in the $3k$-variable reduced-coordinate polynomial ring R_k, cutting out its vanishing locus. We prove, for every $k \geq 2$ over any field of characteristic $\neq 2$, and as four independently-established results rather than one bundled claim: I_k has codimension 2; I_k has exactly 3 minimal generators; R_k/I_k is Cohen-Macaulay; and I_k is radical. The last of these, together with an explicit component count resting on a non-containment argument, assembles into the **Primary Decomposition Theorem**: $I_k = P_2 \cap \cdots \cap P_k$ is an irredundant primary decomposition into exactly $k-1$ primes, following an explicit recursive block-involvement pattern. The proof identifies I_k as the ideal of 2×2 minors of an explicit 2×3 matrix (a determinantal ideal, not merely determinantal-**looking**), and invokes classical determinantal-ideal theory (Bruns-Vetter) and an explicit rank-2 Jacobian witness on every component (Serre's criterion) for the algebraic and radicality halves respectively.

1. Introduction

2×2 matrices A, B have commutator $[A, B] = AB - BA$, and the **iterated** left-normed commutator of k matrices is $[A_1, ..., A_k] := [...[[A_1, A_2], A_3], ..., A_k]$, matching the standard lower-central-series convention for Lie algebras. This object arose out of

- a study of the corresponding formal word in the theory of fiber-counting of word-maps $\pi : F_n \to G$, where F_n is the free group on n letters; in particular, after a close reading of Tambour's result [13]
- a question in algebraic complexity theory – how many scalar multiplications are needed to **compute** $[A_1, ..., A_k]$? – which remains genuinely open for $k \geq 3$ and which the author is presently investigating.

The one fact from that story needed here is purely algebraic, not complexity-theoretic: an explicit bilinear formula for the ordinary 2×2 commutator, chained across k matrices, gives an explicit polynomial parametrization of $[A_1, ..., A_k]$'s reduced coordinates (Definition 3 below), and it is the **vanishing locus** of this parametrization, as an algebraic variety and its coordinate ring, that is the actual subject of this article. One naming note: one could alternatively refer to the coordinate ring of the variety here studied as the "ring-of-k-commuting-2x2-matrices", using "commuting" in this generalized iterated-commutator-vanishing sense. This means that $[A_1, ..., A_k] = 0$, and not literally "every pair of $A_1, ..., A_k$ commutes". The two concepts agree at $k = 2$, where this article's locus **is** the classical commuting variety; for $k \geq 3$ the vanishing of the iterated bracket is the **weaker** of the two conditions, so the locus studied here properly contains the pairwise-commuting one.

The classical commuting variety. For A, B arbitrary (not necessarily 2×2) $n \times n$ matrices, the variety $C_n := \{(A, B) \in M_n \times M_n : AB = BA\}$ – the ordinary, unreduced, un-iterated case this article's own $k = 2$ specializes to – has been studied since the 1950s. Its irreducibility was proven independently by Motzkin and Taussky [3] and by Gerstenhaber [5], the latter also establishing the now-standard bound $\dim_{\mathbb{F}} \mathbb{F}[A, B] \leq n$ for the subalgebra generated by two commuting matrices. Whether the coordinate ring $\mathbb{F}[A, B]/I$, I the ideal generated by the entries of $AB - BA$, is **reduced** and **Cohen-Macaulay** for every n is a much harder question, attributed to Michael Artin and Melvin Hochster – an unpublished, informally-circulated conjecture with no single dateable original source [6], stated precisely as "Conjecture 1.1" in Majidi-Zolbanin and Snapp [7]. Chau [8] records a more specifically dated 1984 conjecture, attributed to Hochster alone and asking only for **reducedness**, about these same commuting loci – settled at $n = 3$ by Thompson shortly afterwards, and open for $n \geq 4$. Freyja Hreinsdóttir's manuscript [9], building on her 1997 Stockholm doctoral thesis under Jan-Erik Roos, attacks the Artin-Hochster conjecture case by case at $n = 3, 4$: syzygies, Betti numbers, the canonical module, and a Koszul-dual structure theorem, verified computationally through $n = 7$ but proven in general only in these small cases. Knutson [10] studies the minimal primes of closely related schemes from a different angle. The state of play for the **classical** commuting variety at the time of writing is therefore: Cohen-Macaulayness is

- settled by hand at $n \leq 2$;
- settled by machine computation at $n = 3$ and $n = 4$, as both Majidi-Zolbanin and Snapp [7] and Hreinsdóttir [9] record; and

- open for every $n \geq 5$.

Reducedness is settled for $n \leq 3$ and open for $n \geq 4$ [8].

At $n = 2$ specifically – this article's $k = 2$ base case – the question is comparatively easy, and is in effect answered again, as a special case, by Theorem 31 below: I_2 (Definition 3) is literally the ideal of the $n = 2$ commuting variety, expressed in reduced (trace-removed) coordinates rather than the full matrix-entry coordinates the classical literature typically uses. The genuinely new object in this article is the **iterated** generalization to $k \geq 3$: not the classical commuting variety at all (see the naming note above), but a structurally analogous determinantal ideal, built by chaining the same bilinear cross-product formula that governs the $n = 2$ case. It is exactly the same **flavor** of question the Artin-Hochster conjecture asks. Is this natural, geometrically meaningful ideal Cohen-Macaulay? Is it reduced? Here those two questions are asked of a different, recursively-defined family of ideals, and this article answers them completely and unconditionally, for every $k \geq 2$ (Theorem 44), rather than case by case or conjecturally. In that sense the results below can be read as a fully worked structural analogue of the still-open classical question, in a setting simple enough ($n = 2$, but k unbounded) to admit a complete answer.

2. Notation

We present the notation used across more than one later result, in order of first appearance. Symbols local to a single proof are omitted. Throughout, $\mathbb{F}$ denotes the ambient field; it is fixed in Section 3 to be of characteristic $\neq 2$, and nothing in this section depends on that.

The following typographic convention is used throughout. An **overline** denotes a **vector** – an element of $\mathbb{F}^3$ (or, for R_k itself, of the free module R_k^3) – while the **same letter, plain, with a two-index comma subscript**, denotes one of that vector's three individual scalar **components**. So $\overline{v}_i \in \mathbb{F}^3$ is block i's whole reduced-coordinate vector, while its three components are the plain scalars $v_{i,1}, v_{i,2}, v_{i,3} \in \mathbb{F}$ (equivalently $v_{i,j}$ for $j = 1, 2, 3$) – never overlined, since a single scalar entry is not itself a vector. The same convention applies to $\overline{w}_j$ versus its components $w_{j,1}, w_{j,2}, w_{j,3}$.

Symbol	Meaning	Defined in
$\overline{v}$ (also $\overline{v}_i$; components $v_{i,1}, v_{i,2}, v_{i,3}$, or $v_{i,j}$)	reduced-coordinate vector in $\mathbb{F}^3$, and its blocks	§3, Setup
S	the sign twist $S = \mathrm{diag}(-1, -1, 1)$, used both as a map and as a matrix	§3, Setup (Theorem 1)
$\mathbb{F}$	ambient field, characteristic $\neq 2$; $\overline{\mathbb{F}}$ a fixed algebraic closure	§3, opening paragraph; $\overline{\mathbb{F}}$ at Remark 43
R_k	the polynomial ring $\mathbb{F}[\overline{v}_1, \ldots, \overline{v}_k]$	Definition 3
I_k	the defining ideal, $I_k \subset R_k$	Definition 3
$\overline{w}_j$	(a) the recursive vector of **polynomials**, an element of R_j^3 (its **evaluations** lie in $\mathbb{F}^3$); (b) as a polynomial map $\mathbb{A}^{3j} \to \mathbb{A}^3$	(a) Definition 3; (b) Lemma 8
U_j	the Zariski-open locus where $\overline{w}_j \neq 0$	Lemma 8
π	the projection $\mathbb{A}^{3k} \to \mathbb{A}^{3(k-1)}$ forgetting $\overline{v}_k$	Definition 14
Φ	the map $\mathbb{A}^{3(k-1)} \times \mathbb{A}^1 \to \mathbb{A}^{3k}$	Definition 14
Y_1, Y_2	the two strata of $V(I_k)$	Definition 14
P_j $(j = 2, \ldots, k)$	the primary components of I_k	Definition 24
J_k	the Jacobian of $\overline{w}_k$	Lemma 37
$[u]_\times$	skew matrix with $[u]_\times v = u \times v$	Lemma 37

3. Setup: the signed encoding and the defining ideal

Throughout, $\mathbb{F}$ is a field of characteristic $\neq 2$, and all matrices, vectors and polynomial rings are over $\mathbb{F}$ unless said otherwise. (Where exactly that characteristic hypothesis is spent is a question in its own right, taken up in Problem 5 at the end of this section; the displayed examples below are computed over $\mathbb{Q}$, but nothing in the definitions is special to it.)

Step 1: discard the scalar part. Write $A = \left(\begin{smallmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{smallmatrix}\right)$ as the sum of its scalar part $(\mathrm{tr}(A)/2)I$ and its traceless remainder

$$A_0 := A - (\mathrm{tr}(A)/2)I = \begin{pmatrix} d_a/2 & a_{12} \\ a_{21} & -d_a/2 \end{pmatrix}, \quad d_a := a_{11} - a_{22}. \tag{1}$$

Scalar matrices are central, so they never contribute to a commutator:

$$[cI, B] = cIB - BcI = cB - cB = 0 \tag{2}$$

for every B and every scalar c, hence $[A, B] = [A_0, B_0]$ identically – no information relevant to **any** bracket is lost by discarding the scalar part, so from here on only the traceless remainder matters.

Step 2: encode the traceless remainder. A traceless matrix $\left(\begin{smallmatrix} x & y \\ z & -x \end{smallmatrix}\right)$ has three free entries. Rather than reading them off directly (x, y, z themselves – shown in Remark 2 below to be a strictly **worse** choice, forcing a swap **and** two separate factors of 2), take

$$\mathrm{enc}(A) := (a_{11} - a_{22}, a_{12} + a_{21}, a_{12} - a_{21}) \in \mathbb{F}^3, \tag{3}$$

the traceless $(1,1)$-entry doubled. (On the traceless part $a_{22} = -a_{11}$, so $d_a = a_{11} - a_{22} = 2a_{11}$: the quantity d_a of Step 1 is literally twice that entry.) The remaining two coordinates are A's two off-diagonal entries replaced by their sum and difference. As with the naive encoding red of Remark 2, no division is needed to **compute** enc – only its inverse (recovering a_{12}, a_{21} individually as $\frac{v_2+v_3}{2}$ and $\frac{v_2-v_3}{2}$) needs 2 invertible, which is the standing hypothesis the article already carries, not a new one.

Step 3: form the commutator. Given traceless $A_0 = \left(\begin{smallmatrix} x & y \\ z & -x \end{smallmatrix}\right)$ and $B_0 = \left(\begin{smallmatrix} u & v \\ w & -u \end{smallmatrix}\right)$, the ordinary matrix commutator $[A_0, B_0] = A_0B_0 - B_0A_0$ is again traceless automatically, since $\mathrm{tr}(PQ) = \mathrm{tr}(QP)$ for any P, Q and therefore $\mathrm{tr}([P, Q]) = 0$ always, so it is again encodable by enc.

Step 4: the encoded bracket is the cross product, up to one fixed sign twist.

For the following theorem, the article's standing hypothesis char $\mathbb{F} \neq 2$ is deliberately not used: the identity holds over every commutative ring.

Theorem 1 (the basic identity)

For traceless 2×2 matrices A_0, B_0 over $\mathbb{F}$, with no hypothesis on the characteristic,

$$\mathrm{enc}([A_0, B_0]) = S(\mathrm{enc}(A_0) \times \mathrm{enc}(B_0)), \quad S := \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{4}$$

Proof of Theorem 1. A direct computation, carried out symbolically over generic entries $a_{11}, \ldots, b_{22}$ (not merely checked at sampled points) in *code/verify_lorentzian_encoding_identity.sage*: both sides, expanded in the 8 free entries, agree term by term. Both sides are integer polynomials in those entries, so the identity descends to every commutative ring; it was checked over $\mathbb{Z}$ and, separately, over $\mathbb{F}_2$ – where it still holds – precisely to confirm that this step is not one of the places where char $\mathbb{F} \neq 2$ is spent (see Problem 5). □

Remark 2 (why this particular twist, and not some other one)

There are three concrete respects in which S improves on the naive reshuffle $\tau(a,b,c) = (b,a,2c)$ that the encoding $\mathrm{red}(A) := (a_{12}, a_{21}, d_a)$ forces, each checked, not merely asserted:

1. **S does not mix coordinates.** τ swaps two of its three inputs; S acts diagonally, one coordinate at a time. Consequence: red would need a page-length remark ("why $\overline{w}_j$ carries no outer τ") to justify that rescaling+swapping a generating set by an invertible linear map doesn't change the ideal generated. Here the analogous fact is immediate: **individually** negating two of three generators obviously leaves the ideal unchanged, generator by generator – no swap to account for. (Still verified computationally, not merely asserted: $\mathrm{ideal}(\overline{w}_2) = \mathrm{ideal}(S\overline{w}_2)$, confirmed by direct Gröbner-basis equality in *code/verify_encoding_gl3_equivariance.sage*.)
2. **S is an involution; τ is not.** $S^2 = I$ exactly (immediate from the diagonal entries being ± 1), while $\tau^2 = \mathrm{diag}(1,1,4) \neq I$.
3. **S is not an arbitrary matrix that happens to work – it is (up to a constant) the matrix of $\mathfrak{sl}_2$'s own Killing form, in these coordinates.** For traceless $A_0 = \left(\begin{smallmatrix} x & y \\ z & -x \end{smallmatrix}\right)$, $\det(A_0) = -x^2 - yz$, an **indefinite** quadratic form (never definite: $-yz = -((y+z)/2)^2 + ((y-z)/2)^2$ is a genuine difference of squares whenever $2 \neq 0$). Checked directly: $-\mathrm{enc}(A)_1^2 - \mathrm{enc}(A)_2^2 + \mathrm{enc}(A)_3^2 = 4\det(A_0)$, exactly (Sage, symbolic). So S's sign pattern **is** det's own signature, read off in these coordinates – not a bookkeeping accident. (Lie-theoretically: $\mathfrak{sl}_2(\mathbb{R}) \cong \mathfrak{so}(2,1)$, the Lorentz algebra, not the compact $\mathfrak{so}(3)$ one gets only after adjoining i – classical, not cited to a specific source here, the same way this article leaves certain folklore facts uncited rather than inventing a citation.) Over $\mathbb{R}$, or over any formally real field such as $\mathbb{Q}$, that sign cannot be removed by **any** invertible linear re-encoding: $-x^2 - y^2 + z^2$ is isotropic, vanishing at $(1,0,1)$, whereas $x^2 + y^2 + z^2$ is anisotropic, since over a formally real field a sum of squares vanishes only when every term does – and isotropy is manifestly preserved by invertible linear substitution. (This is the content of Sylvester's law of inertia in the case at hand; the law itself is a statement about $\mathbb{R}$.) In that precise sense S's single-sign twist is already the simplest one achievable, not merely simpler than τ by luck.

   Over a general field the claim genuinely fails, and it fails already in the smallest interesting case, so the restriction above is not idle caution: over $\mathbb{F}_3$, where -1 is **not** a square, $\mathrm{diag}(-1,-1,1)$ is nevertheless congruent to the identity form, via $T = \left(\begin{smallmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 2 & 0 \end{smallmatrix}\right)$ (of determinant 1) with $T^\top T = \mathrm{diag}(2,2,1) = \mathrm{diag}(-1,-1,1)$. So "$-1$ is not a square" is **not** the right criterion; formal reality is.

The "obvious" un-permuted reading of the matrix entries, $\mathrm{raw}(A) := (x,y,z)$, is worse than either encoding: the twist it forces is $\mathrm{swap}(2,3) \cdot \mathrm{diag}(1,2,2)$ – a coordinate swap **and** two separate factors of 2. A different escape, adjoining i to get a literal untwisted cross product ($\mathfrak{su}(2) \cong \mathfrak{so}(3)$ in place of $\mathfrak{so}(2,1)$), is a genuine alternative but a strictly narrower one, since it constrains the field.

◇

Definition 3 (reduced iterated commutator, signed encoding)

$$\overline{w}_2 := \overline{v}_1 \times \overline{v}_2, \quad \overline{w}_j := S(\overline{w}_{j-1}) \times \overline{v}_j \quad (j = 3, \ldots, k). \tag{5}$$

Let $R_k := \mathbb{F}[\overline{v}_1, \ldots, \overline{v}_k]$ ($\overline{v}_i \in \mathbb{F}^3$, $3k$ variables) and $I_k \subset R_k$ the ideal generated by the three components of $\overline{w}_k$ – the ideal cutting out $[A_1, \ldots, A_k] = 0$ in reduced coordinates. Two deliberate understatements here. First, I_k is at this point only the **generated** ideal; that it is also the full vanishing ideal of that locus is Theorem 42, in §7 below, and is not to be assumed before then. Second, the encoded bracket is strictly $S(\overline{w}_k)$, not $\overline{w}_k$ – but S merely negates two of three coordinates, so the two triples generate the same ideal, and it is cleaner to carry the untwisted one.

♣

To spell out exactly what this means: "$R_k := \mathbb{F}[\overline{v}_1, \ldots, \overline{v}_k]$" is shorthand for **the polynomial ring in the $3k$ scalar variables $v_{i,j}$ ($i = 1, \ldots, k$, $j = 1, 2, 3$)**, grouped into k blocks of 3 – **not** a ring with k vector-valued generators (a polynomial ring's generators are always scalars). At $k = 2$, written out in full:

$$R_2 = \mathbb{F}\left[v_{1,1}, v_{1,2}, v_{1,3}, v_{2,1}, v_{2,2}, v_{2,3}\right], \tag{6}$$

six ordinary scalar indeterminates, with $\overline{v}_1 = (v_{1,1}, v_{1,2}, v_{1,3})$ and $\overline{v}_2 = (v_{2,1}, v_{2,2}, v_{2,3})$ merely a notational grouping of them into two blocks of three – convenient because $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2$ is most naturally written as a cross product of two block-vectors, not as some formula in six unrelated scalars.

Each $\overline{w}_j$ is a triple of polynomials, i.e. $\overline{w}_j \in R_j^3$ (the free rank-3 module over R_j) – equivalently, exactly as recorded in the Notation table's entry (b) for $\overline{w}_j$, the polynomial map $\overline{w}_j : \mathbb{A}^{3j} \to \mathbb{A}^3$ **given by substitution**: evaluating each of $\overline{w}_j$'s three polynomial components at a point $(\overline{v}_1, \ldots, \overline{v}_j) \in \mathbb{A}^{3j}$. Concretely at $k = 2$, writing out all six variables, $\overline{w}_2$ is the substitution map

$$\begin{pmatrix} v_{1,1} \\ v_{1,2} \\ v_{1,3} \\ v_{2,1} \\ v_{2,2} \\ v_{2,3} \end{pmatrix} \in \mathbb{A}^6 \quad \mapsto \quad \begin{pmatrix} v_{1,2}v_{2,3} - v_{1,3}v_{2,2} \\ v_{1,3}v_{2,1} - v_{1,1}v_{2,3} \\ v_{1,1}v_{2,2} - v_{1,2}v_{2,1} \end{pmatrix} \in \mathbb{A}^3, \tag{7}$$

and I_k (Definition 3) is, equivalently, the ideal generated by the three coordinate functions of $\overline{w}_k : \mathbb{A}^{3k} \to \mathbb{A}^3$ pulled back along this substitution.

Example 4 ($\overline{w}_k$ and I_k, explicitly, for $k = 2, 3, 4$)

Computed directly, not hand-derived. At $k = 2$, the three components of $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2$:

$$v_{1,2}v_{2,3} - v_{1,3}v_{2,2}, \quad v_{1,3}v_{2,1} - v_{1,1}v_{2,3}, \quad v_{1,1}v_{2,2} - v_{1,2}v_{2,1}. \tag{8}$$

(This is what red would give at this stage too – the two encodings agree exactly at the very first stage, since neither τ nor S is applied yet.) At $k = 3$, the three components of $\overline{w}_3 = S(\overline{w}_2) \times \overline{v}_3$ – now with **every coefficient equal to ± 1**, no factor of 2 anywhere, unlike the τ-twisted $\overline{w}_3$ that red produces:

$$v_{1,2}v_{2,1}v_{3,2} - v_{1,1}v_{2,2}v_{3,2} - v_{1,3}v_{2,1}v_{3,3} + v_{1,1}v_{2,3}v_{3,3}, \tag{9}$$

$$-v_{1,2}v_{2,1}v_{3,1} + v_{1,1}v_{2,2}v_{3,1} - v_{1,3}v_{2,2}v_{3,3} + v_{1,2}v_{2,3}v_{3,3}, \tag{10}$$

$$v_{1,3}v_{2,1}v_{3,1} - v_{1,1}v_{2,3}v_{3,1} + v_{1,3}v_{2,2}v_{3,2} - v_{1,2}v_{2,3}v_{3,2}, \tag{11}$$

and $I_3 := (w_{3,1}, w_{3,2}, w_{3,3})$. At $k = 4$, $\overline{w}_4 = S(\overline{w}_3) \times \overline{v}_4$: again every coefficient is ± 1; the first component of $\overline{w}_4$ is

$$-v_{1,3}v_{2,1}v_{3,1}v_{4,2} + v_{1,1}v_{2,3}v_{3,1}v_{4,2} - v_{1,3}v_{2,2}v_{3,2}v_{4,2} + v_{1,2}v_{2,3}v_{3,2}v_{4,2} \tag{12}$$

$$+v_{1,2}v_{2,1}v_{3,1}v_{4,3} - v_{1,1}v_{2,2}v_{3,1}v_{4,3} + v_{1,3}v_{2,2}v_{3,3}v_{4,3} - v_{1,2}v_{2,3}v_{3,3}v_{4,3}, \tag{13}$$

the other two of the same shape (omitted for space; all three components of $\overline{w}_4$, and $I_4 := (w_{4,1}, w_{4,2}, w_{4,3})$, in *code/compute_signed_encoding_examples.sage*'s output). ◇

The displayed examples above were computed over $\mathbb{Q}$, but every result below holds over the arbitrary field $\mathbb{F}$ of characteristic $\neq 2$ fixed at the start of this section. Where that hypothesis is actually spent is not where one first expects, and the answer leaves a question we are happy to hand on:

Problem 5 (characteristic 2)

The hypothesis is **not** spent on S. Since $\det S = 1$, the twist is invertible over every field, and Remark 2′s "an invertible change of generators does not change the ideal" needs no hypothesis whatsoever. Under the naive reshuffle it would: $\det \tau = -2$, so τ is singular in characteristic 2 precisely.

It is spent in **Step 1**, earlier than Step 2′s division by 2 and more seriously. In characteristic 2 one has $\operatorname{tr}(cI) = 2c = 0$, so **scalar matrices are themselves traceless**. The splitting $A = (\operatorname{tr}(A)/2)I + A_0$ then has no meaning at all: the scalars do not sit beside a complement, they sit **inside** the traceless matrices.

The bracket is unbothered — scalars are central in every characteristic — so $[\cdot,\cdot]$ still descends to the three-dimensional quotient $\mathfrak{gl}_2/(\mathbb{F}\cdot I)$. The question is whether the rest of the article descends with it.

Question. Over a field of characteristic 2, is there a linear isomorphism $\mathfrak{gl}_2/(\mathbb{F}\cdot I) \cong \mathbb{F}^3$ carrying the induced bracket to a twisted cross product $E(\cdot\times\cdot)$ for some invertible E? And if so, do Theorem 19, Theorem 12, Theorem 31, Theorem 42 and Theorem 44 hold verbatim?

What makes this more than idle curiosity: **every polynomial displayed in Example 4 has coefficients ± 1 only**. No 2 appears anywhere in the output. Whatever the obstruction is, it lives entirely in the setup and not in the ideal — which is exactly the situation in which a different setup might dissolve it.

Computational evidence, bearing on the second half of the question only. Nothing in Definition 3′s recursion needs a characteristic hypothesis: $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2$ and $\overline{w}_j = S(\overline{w}_{j-1}) \times \overline{v}_j$ define an ideal $I_k \subset R_k$ over **any** field. Taking that recursion as the definition and running the four results in characteristic 2 anyway, for $k = 2,\dots,5$ over $\mathbb{F}_2$ and $\mathbb{F}_4$: the codimension is 2, the three generators are linearly independent over the base field, I_k is radical, and $\operatorname{pd}(R_k/I_k) = 2$, so R_k/I_k is Cohen-Macaulay — in every case, with no deviation whatever from the characteristic-zero behaviour (*code/verify_over_finite_fields.sage* and *code/verify_cm_over_finite_fields.m2* the same runs also cover $\mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7, \mathbb{F}_9$ and $\mathbb{Q}$). The **conclusions**, then, appear to survive characteristic 2 untouched — which is evidence for the paragraph above, not against it.

What no computation of this kind can settle is the **first** half of the question: whether a twisted-cross-product model of $\mathfrak{gl}_2/(\mathbb{F}\cdot I)$ exists there at all, so that this I_k is once again the ideal of an iterated-commutator locus rather than merely a recursion that happens to go on behaving well. That is where the exercise lies.

4. The determinantal structure and the generator count

Theorem 6 (determinantal structure)

For $k \geq 3$, I_k is the ideal of 2×2 minors of the 2×3 matrix

$$\begin{pmatrix} S(\overline{w}_{k-1}) \\ \overline{v}_k \end{pmatrix}, \tag{14}$$

i.e. $V(I_k)$ is the preimage, under $(\overline{v}_1,\dots,\overline{v}_k) \mapsto \begin{pmatrix} S(\overline{w}_{k-1}(\overline{v}_1,\dots,\overline{v}_{k-1})) \\ \overline{v}_k \end{pmatrix}$, of the classical rank-$\leq 1$ determinantal variety of 2×3 matrices. (At $k = 2$ the same statement holds with $S(\overline{w}_{k-1})$ replaced by $\overline{v}_1$ itself, matching Definition 3′s base case $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2$ directly.)

Proof of Theorem 6. The claim is immediate from two facts taken together. The three components of a cross product $a \times b$ are, up to sign, the 2×2 minors of the 2×3 matrix with rows a and b; and Definition 3′s recursion builds $\overline{w}_k$ from exactly such a cross product. The statement was also checked independently at $k = 2,\dots,6$ in *code/verify_determinantal_structure_all_k.sage*, in two separate ways. First, I_k and the ideal of the displayed minors are shown equal by reducing each ideal's generators to zero against the other's Gröbner basis, in **both** directions, rather than by containment one way. Second – and this is what makes "independent" mean something here – the recursion itself is checked against honest 2×2 matrix multiplication: the A_i are reconstructed from their enc-coordinates, the iterated commutator is formed by repeated matrix products, and the result encoded and compared against $S(\overline{w}_k)$. The two routes share no code. The underlying polynomial identity $\operatorname{enc}([A,B]) = S(\operatorname{enc}(A)\times \operatorname{enc}(B))$ of Theorem 1 makes this a two-line induction valid for **every** k, not merely a pattern checked at small k (*code/verify_lorentzian_encoding_identity.sage*). □

This theorem is the engine behind the geometric results of §5: R_k/I_k is not merely **determinantal-looking**, it is literally an iterated extension of the coordinate ring of a classical determinantal variety.

Example 7 (the 2×3 matrix of Theorem 6, for $k = 2, 3, 4$)

At $k = 2$:

$$\begin{pmatrix} \overline{v}_1 \\ \overline{v}_2 \end{pmatrix} = \begin{pmatrix} v_{1,1} & v_{1,2} & v_{1,3} \\ v_{2,1} & v_{2,2} & v_{2,3} \end{pmatrix}. \tag{15}$$

At $k = 3$, the first row is $S(\overline{w}_2) = (-w_{2,1}, -w_{2,2}, w_{2,3})$ where $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2$ is as in the previous example, giving

$$\begin{pmatrix} S(\overline{w}_2) \\ \overline{v}_3 \end{pmatrix} = \begin{pmatrix} v_{1,3}v_{2,2} - v_{1,2}v_{2,3} & -v_{1,3}v_{2,1} + v_{1,1}v_{2,3} & -v_{1,2}v_{2,1} + v_{1,1}v_{2,2} \\ v_{3,1} & v_{3,2} & v_{3,3} \end{pmatrix}. \tag{16}$$

At $k = 4$ the first row is $S(\overline{w}_3)$, whose three entries are cubic with four terms each. Naming them A, B, C keeps the matrix legible:

$$\begin{pmatrix} A & B & C \\ v_{4,1} & v_{4,2} & v_{4,3} \end{pmatrix}, \tag{17}$$

where

$$\begin{aligned} A &= -v_{1,2}v_{2,1}v_{3,2} + v_{1,1}v_{2,2}v_{3,2} + v_{1,3}v_{2,1}v_{3,3} - v_{1,1}v_{2,3}v_{3,3}, \\ B &= v_{1,2}v_{2,1}v_{3,1} - v_{1,1}v_{2,2}v_{3,1} + v_{1,3}v_{2,2}v_{3,3} - v_{1,2}v_{2,3}v_{3,3}, \\ C &= v_{1,3}v_{2,1}v_{3,1} - v_{1,1}v_{2,3}v_{3,1} + v_{1,3}v_{2,2}v_{3,2} - v_{1,2}v_{2,3}v_{3,2}. \end{aligned} \tag{18}$$

The row degree grows by one at each step $(1, 2, 3, \ldots)$, while the second row is always the fresh, free vector $\overline{v}_k$ – exactly the "one fixed complicated row, one free row" shape Proposition 29 needs. ◇

Lemma 8 (non-degeneracy)

For every $j \geq 2$, the polynomial map $\overline{w}_j : \mathbb{A}^{3j} \to \mathbb{A}^3$ is not identically zero. Hence $U_j := \{p \in \mathbb{A}^{3j} : \overline{w}_j(p) \neq 0\}$ is a nonempty Zariski-open subset. ♡

Proof of Lemma 8. By induction: $\overline{w}_2(e_1, e_2) = e_1 \times e_2 = e_3 \neq 0$; given $c := \overline{w}_{j-1}(p) \neq 0$, at least one standard basis vector is not proportional to $S(c)$, and a cross product vanishes only on linearly dependent pairs, so $\overline{w}_{j(p,e_i)} = S(c) \times e_i \neq 0$ for a suitable e_i. □

The generator count of I_k needs nothing beyond the definition and non-degeneracy – not codimension, not the determinantal-structure theorem above, not any external citation:

Lemma 9 (independent same-degree generators are minimal)

If a homogeneous ideal in a graded ring is generated by finitely many forms of the same degree that are linearly independent over the base field, that generating set is minimal. ♡

Proof of Lemma 9. A polynomial-coefficient combination reproducing one of the generators from the others, while staying in the same degree, can only use degree-0 (constant) coefficients – so a redundant generator would violate linear independence directly. □

Lemma 10 (the three generators of I_k are independent)

The three components of $\overline{w}_k$ are linearly independent over $\mathbb{F}$, for every $k \geq 2$. ♡

Proof of Lemma 10. By induction on k, and the induction hypothesis must be **independence** itself; see Remark 11 for why the weaker non-degeneracy of Lemma 8 does not suffice.

Base case $k = 2$. The three components of $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2$ are the 2×2 minors of the generic 2×3 matrix with rows $\overline{v}_1, \overline{v}_2$, displayed in Example 4. Each contains a monomial the other two do not — $v_{1,2}v_{2,3}$ occurs only in the first, $v_{1,3}v_{2,1}$ only in the second, $v_{1,1}v_{2,2}$ only in the third — so no nontrivial constant relation is possible.

Induction step. Suppose the components of $\overline{w}_{k-1}$ are linearly independent over $\mathbb{F}$, and let $c \in \mathbb{F}^3$ satisfy $c \cdot \overline{w}_k = 0$. Writing $a := S(\overline{w}_{k-1})$, the scalar triple product identity gives

$$0 = c \cdot (a \times \overline{v}_k) = (c \times a) \cdot \overline{v}_k. \tag{19}$$

The variables $\overline{v}_k$ are a **fresh** block, occurring in neither a nor c, and the right-hand side is linear in them; so each of its three coefficients vanishes separately, i.e.

$$c \times a = 0. \tag{20}$$

Suppose $c \neq 0$. Then $c \times a = 0$ forces $a = f \cdot c$ for a single polynomial f — over the fraction field $c \times a = 0$ says a lies in the line spanned by c, and $f = a_i/c_i$ is a polynomial for any index with $c_i \neq 0$. But then a's three components are pairwise proportional, hence linearly **dependent** over $\mathbb{F}$ (any d with $d \cdot c = 0$ gives $d \cdot a = (d \cdot c)f = 0$, and such a nonzero d exists since c spans only a line in $\mathbb{F}^3$). Since S is invertible, the components of $\overline{w}_{k-1} = S^{-1}(a)$ are then linearly dependent too, contradicting the induction hypothesis. Hence $c = 0$.

Corroborated computationally: the coefficient matrix has rank 3 at $k = 2, \ldots, 6$ (*code/verify_generator_independence.sage*). □

Remark 11 (why non-degeneracy is not enough here)

It is tempting to argue instead that a constant relation would force $S(\overline{w}_{k-1})$ to be a **constant vector**, and to rule that out by Lemma 8. That argument is not valid. The relation forces only $c \times S(\overline{w}_{k-1}) = 0$, which makes the components of $S(\overline{w}_{k-1})$ pairwise proportional — not constant. A vector such as $(f, 2f, 3f)$ with $f \neq 0$ is nonvanishing, so Lemma 8 does not exclude it, yet its components are visibly dependent. Non-degeneracy is strictly weaker than what the step needs, which is why the induction above carries linear independence itself.

Theorem 12 (I_k has exactly 3 minimal generators)

For every $k \geq 2$: I_k has exactly 3 minimal generators.

Proof of Theorem 12. Immediate from Lemma 9 applied to Lemma 10's independence, since I_k is generated by the three (same-degree-k-homogeneous) components of $\overline{w}_k$ by definition. □

5. The geometric structure of $V(I_k)$

Lemma 13 (continuous images and closures preserve irreducibility)

Let $f : X \to Y$ be a continuous map of topological spaces with X irreducible. Then $f(X)$ (with the subspace topology) is irreducible, and hence so is $\overline{f(X)}$.

Proof of Lemma 13. **Image.** Suppose $f(X) = C_1 \cup C_2$ with C_1, C_2 closed in $f(X)$. By continuity, $f^{-1}(C_1)$ and $f^{-1}(C_2)$ are closed in X, and $X = f^{-1}(C_1) \cup f^{-1}(C_2)$. Since X is irreducible, one of the two equals X – say $f^{-1}(C_1) = X$ – so $f(X) \subseteq C_1$, i.e. $C_1 = f(X)$, not proper. Hence $f(X)$ is irreducible. **Closure.** If $A \subseteq Y$ is irreducible and $\overline{A} = C_1 \cup C_2$ with C_1, C_2 closed in Y, then $A = (A \cap C_1) \cup (A \cap C_2)$; irreducibility of A forces, say, $A \subseteq C_1$; since C_1 is closed, $\overline{A} \subseteq C_1$, and combined with $C_1 \subseteq \overline{A}$, $C_1 = \overline{A}$, not proper. Apply both parts with $A := f(X)$. □

This is purely topological – continuity of f is the only hypothesis used, nothing about f being a polynomial map or morphism of varieties [11].

Definition 14 (the strata Y_1, Y_2)

For $k \geq 3$, write π for the projection **forgetting $\overline{v}_k$**,

$$\pi : \mathbb{A}^{3k} \to \mathbb{A}^{3(k-1)}, \quad \pi(\overline{v}_1, \ldots, \overline{v}_k) := (\overline{v}_1, \ldots, \overline{v}_{k-1}), \tag{21}$$

and Φ for the map

$$\Phi : \mathbb{A}^{3(k-1)} \times \mathbb{A}^1 \to \mathbb{A}^{3k}, \quad \Phi(p, \lambda) := (p, \lambda S(\overline{w}_{k-1}(p))). \tag{22}$$

Define two subsets of $\mathbb{A}^{3k}$:

$$Y_1 := V(I_{k-1}) \times \mathbb{A}^3, \quad Y_2 := \overline{\Phi(\mathbb{A}^{3k-2})}. \tag{23}$$

Y_1 is the locus where the previous bracket already vanishes, with $\overline{v}_k$ completely free; Y_2 is the closure of the locus swept out as $\overline{v}_k$ ranges over scalar multiples of $S(\overline{w}_{k-1}(p))$, over all $p \in \mathbb{A}^{3(k-1)}$. At $k = 2$ there is no "previous bracket" to chain from, so neither Y_1 nor Φ is defined; by convention $Y_2 := V(I_2)$ directly there, matching Theorem 6's own base-case treatment.

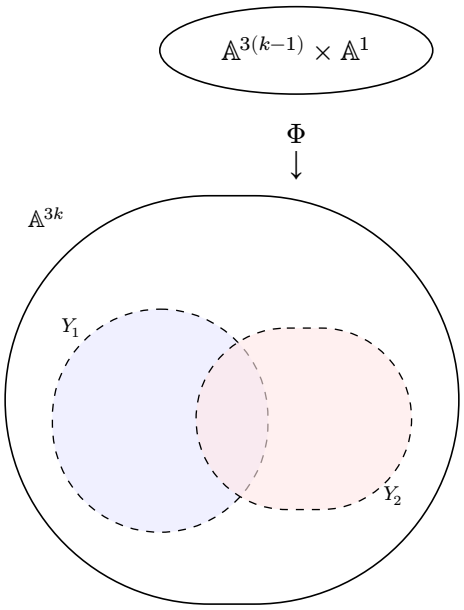


Figure 1: Both strata of $V(I_k) = Y_1 \cup Y_2$ inside the same ambient $\mathbb{A}^{3k}$: $Y_1 := V(I_{k-1}) \times \mathbb{A}^3$ (identifying $\mathbb{A}^{3(k-1)} \times \mathbb{A}^3 \cong \mathbb{A}^{3k}$) and $Y_2 := \overline{\Phi(\mathbb{A}^{3k-2})}$, Φ's image (Φ shown mapping in from above, into Y_2 specifically – Y_1 is not part of Φ's image). The two overlap (not disjoint, Proposition 21 and Proposition 22 rule out either containing the other): $\Phi(p, 0) = (p, 0)$ for **every** p, so the whole zero-section $\{(p, 0)\}$ lies in Y_2, and its intersection with Y_1 is $V(I_{k-1}) \times \{0\}$, of dimension $3k - 5$ – properly smaller than either stratum's own dimension $3k - 2$.

Lemma 15 (Y_2 is irreducible)

For $k \geq 3$: Y_2 (Definition 14) is irreducible, and in fact its defining ideal (the graph ideal of Φ with λ eliminated) is prime.

Proof of Lemma 15. Φ is a polynomial map, hence continuous for the Zariski topology (preimages of Zariski-closed sets under a polynomial map are again Zariski-closed, by substitution), and its domain $\mathbb{A}^{3k-2}$ is irreducible (the polynomial ring over $\mathbb{F}$ is a domain, so the zero ideal is prime). Lemma 13 gives Y_2 irreducible directly. Primality of the elimination ideal follows the same way: it is the kernel of a map into a domain ($\mathbb{F}[\overline{v}_1, \ldots, \overline{v}_k]/(\cdot) \cong \mathbb{F}[\overline{v}_1, \ldots, \overline{v}_{k-1}, \lambda]$), hence prime – this holds for every k by this argument alone, additionally corroborated computationally at $k = 4$ by computing the elimination ideal directly and confirming primality in Sage, as an independent check rather than a substitute for the argument. □

Example 16 (the defining ideals of Y_1 and Y_2, for $k = 2, 3, 4$)

Writing $\overline{v}_i = (v_{i,1}, v_{i,2}, v_{i,3})$ for the three coordinates of block $\overline{v}_i$: at $k = 2$, $Y_2 := V(I_2)$ (the base-case convention above), so its ideal **is** I_2, generated by

$$v_{1,2}v_{2,3} - v_{1,3}v_{2,2}, \quad v_{1,3}v_{2,1} - v_{1,1}v_{2,3}, \quad v_{1,1}v_{2,2} - v_{1,2}v_{2,1}. \tag{24}$$

(Y_1 is not defined at $k = 2$, Definition 14.) For $k \geq 3$, Y_1's defining ideal is simply I_{k-1} itself, viewed inside R_k by leaving $\overline{v}_k$ free – at $k = 3$, exactly the three generators displayed above for I_2, now inside R_3 with $\overline{v}_3$ not appearing at all; at $k = 4$, exactly I_3's three generators (displayed below in this same example) inside R_4 with $\overline{v}_4$ free. Unlike Y_2, no elimination is needed to compute Y_1's ideal – it is already I_{k-1}'s own generating set, unchanged. At $k = 3$, Y_2's elimination ideal (computed directly, not inferred) is generated by

$$v_{2,1}v_{3,1} + v_{2,2}v_{3,2} - v_{2,3}v_{3,3}, \quad v_{1,1}v_{3,1} + v_{1,2}v_{3,2} - v_{1,3}v_{3,3}, \tag{25}$$

$$v_{1,2}v_{2,1}v_{3,2} - v_{1,1}v_{2,2}v_{3,2} - v_{1,3}v_{2,1}v_{3,3} + v_{1,1}v_{2,3}v_{3,3} \tag{26}$$

– and these three repay a careful reading, because their shape is not the one the parametrization first suggests. **All three** are linear in the new block $\overline{v}_3$; what separates them is total degree, $2, 2, 3$. The two quadratic ones are, up to sign, the incidence conditions

$$\overline{v}_1 \cdot S(\overline{v}_3) = 0, \quad \overline{v}_2 \cdot S(\overline{v}_3) = 0. \tag{27}$$

These are **consequences** of the proportionality $\overline{v}_3 \propto S(\overline{w}_2)$, not its components: substituting $\overline{v}_3 = \lambda S(\overline{w}_2)$ and using $S^2 = I$ turns them into $\lambda(\overline{v}_i \cdot \overline{w}_2) = 0$, which holds because $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2$ is orthogonal to both of its factors. (The components of $\overline{v}_3 \times S(\overline{w}_2)$, which **are** the proportionality condition, are cubic, and none of them appears above.) The third generator, of degree 3, is verbatim the first component of $\overline{w}_3$. Note that no generator of this ideal could avoid $\overline{v}_3$ altogether: Y_2 projects dominantly onto $\mathbb{A}^6$, so the elimination ideal meets R_2 in 0. **This ideal is a proper factor of I_3, not equal to it** – I_3 is the intersection of this ideal with a second prime (the "Y_1" piece, where the $k = 2$ bracket already vanishes); see the component-count example after Theorem 23 below, where this exact ideal reappears as one of I_3's two primary components. At $k = 4$ the same pattern repeats one degree higher, the three generators now having degrees $2, 3, 4$ and again all three linear in the new block $\overline{v}_4$. The first two are once more incidence conditions forced by $\overline{w}_3 \perp S(\overline{w}_2), \overline{v}_3$, namely $\overline{v}_3 \cdot S(\overline{v}_4) = 0$ and $\overline{w}_2 \cdot \overline{v}_4 = 0$; the third is the negative of $\overline{w}_4$'s first component:

$$v_{3,1}v_{4,1} + v_{3,2}v_{4,2} - v_{3,3}v_{4,3}, \tag{28}$$

$$v_{1,3}v_{2,2}v_{4,1} - v_{1,2}v_{2,3}v_{4,1} - v_{1,3}v_{2,1}v_{4,2} + v_{1,1}v_{2,3}v_{4,2} + v_{1,2}v_{2,1}v_{4,3} - v_{1,1}v_{2,2}v_{4,3}, \tag{29}$$

$$v_{1,3}v_{2,1}v_{3,1}v_{4,2} - v_{1,1}v_{2,3}v_{3,1}v_{4,2} + v_{1,3}v_{2,2}v_{3,2}v_{4,2} - v_{1,2}v_{2,3}v_{3,2}v_{4,2} \tag{30}$$

$$-v_{1,2}v_{2,1}v_{3,1}v_{4,3} + v_{1,1}v_{2,2}v_{3,1}v_{4,3} - v_{1,3}v_{2,2}v_{3,3}v_{4,3} + v_{1,2}v_{2,3}v_{3,3}v_{4,3}. \tag{31}$$

All computed directly (elimination of λ from the graph ideal of Φ, in Sage), not just asserted from the pattern. ◇

Remark 17 (the closure, not the raw image, is what is needed)

A polynomial map's image need not itself be closed – the standard example is $\mathbb{A}^2 \to \mathbb{A}^2$, $(x, y) \mapsto (x, xy)$, whose image is not even locally closed, though it is **constructible** (a finite union of locally closed sets) by Chevalley's theorem, which has no purely topological analogue for general continuous maps [11]. This never causes a problem for Y_2, because Y_2 is **defined** as $\overline{\Phi(\cdot)}$ from the outset, and the decomposition used in Theorem 19′s proof below, $V(I_k) = Y_1 \cup Y_2$, is established by a direct case split rather than by taking closures and hoping: a point $(p, \overline{v}_k) \in V(I_k)$ either has $S(\overline{w}_{k-1}(p)) = 0$, putting it in Y_1 directly, or has $\overline{v}_k = \lambda S(\overline{w}_{k-1}(p))$ for some λ, so $(p, \overline{v}_k) = \Phi(p, \lambda)$ is **literally** a value of Φ – not merely a limit point – hence in Y_2 with no appeal to closure at all. The reverse containment $Y_1 \cup Y_2 \subseteq V(I_k)$ needs only that $\Phi(\cdot) \subseteq V(I_k)$ (a cross product of a vector with itself vanishes: $S(\overline{w}_{k-1}) \times (\lambda S(\overline{w}_{k-1})) = 0$ identically) together with $V(I_k)$ already being closed, automatic since it is a vanishing locus by definition. So the theorem never needs "the image of Φ is closed" – only that it is contained in the already-closed $V(I_k)$, which is immediate. ◇

Lemma 18 (dimension of Y_2, and where it lives)

With Y_2 as in Lemma 15 and U_{k-1} as in Lemma 8: $\dim Y_2 = 3k - 2$, and Y_2 meets $\pi^{-1}(U_{k-1})$ nonemptily, where π is the projection of Definition 14, forgetting $\overline{v}_k$. ♡

Proof of Lemma 18. $\pi : \mathbb{A}^{3k} \to \mathbb{A}^{3(k-1)}$ discards the last block: $\pi(\overline{v}_1, \ldots, \overline{v}_k) = (\overline{v}_1, \ldots, \overline{v}_{k-1})$. Hence $\pi^{-1}(U_{k-1}) = U_{k-1} \times \mathbb{A}^3$, the set of points whose **first** $k-1$ blocks lie in U_{k-1} and whose $\overline{v}_k$ is completely unconstrained. Over U_{k-1}, Φ is injective. Both p and λ can be recovered from $\Phi(p, \lambda)$: p directly, and λ by dividing any nonzero coordinate of $\overline{v}_k = \lambda S(\overline{w}_{k-1}(p))$ by the corresponding coordinate of $S(\overline{w}_{k-1}(p))$; this division is well-defined since $S(\overline{w}_{k-1}(p)) \neq 0$ on U_{k-1}. Injectivity gives $\dim Y_2 = \dim(U_{k-1} \times \mathbb{A}^1) = 3(k-1) + 1 = 3k - 2$. Since U_{k-1} is nonempty by Lemma 8, $\Phi(U_{k-1} \times \{1\})$ is a nonempty subset of $Y_2 \cap \pi^{-1}(U_{k-1})$. □

Theorem 19 (I_k has codimension 2)

For every $k \geq 2$: $\dim V(I_k) = 3k - 2$, i.e. $\operatorname{codim}(I_k) = 2$.

Proof of Theorem 19. Induction on k. **Base case $k = 2$**: $V(I_2)$ is the rank-≤ 1 locus of a 2×3 matrix with independent-indeterminate entries, of the standard dimension $4 = 3 \cdot 2 - 2$ for the generic determinantal variety of this shape [3]. **Inductive step**: by Theorem 6, and since $a \times b = 0$ if and only if a, b are linearly dependent, $V(I_k) = Y_1 \cup Y_2$ (Definition 14). The locus $S(\overline{w}_{k-1}) = 0$ gives Y_1, with $\overline{v}_k$ unconstrained; the locus $S(\overline{w}_{k-1}) \neq 0$ with $\overline{v}_k$ proportional to it gives a subset of Y_2. This is the direct case split of Remark 17, not merely a closure-level approximation. By the inductive hypothesis, $\dim V(I_{k-1}) = 3(k-1) - 2$, so $\dim Y_1 = 3(k-1) - 2 + 3 = 3k - 2$. Also $\dim Y_2 = 3k - 2$, by Lemma 18. Hence $\dim V(I_k) = \dim(Y_1 \cup Y_2) = 3k - 2$. This dimension statement alone needs no containment argument; that is needed only for the sharper component-count result below. □

Example 20 (dimension and codimension, for $k = 2, 3, 4$)

Values of $\dim R_k$ (the ambient dimension, $= 3k$), $\dim(R_k/I_k)$ (the dimension of the variety $V(I_k)$ – **not** "$\dim I_k$", an ideal is not a variety and does not itself have a dimension in this sense), and $\operatorname{codim}(I_k) := \dim R_k - \dim(R_k/I_k)$:

k	$\dim R_k$	$\dim(R_k/I_k)$	$\operatorname{codim}(I_k)$
2	6	4	2
3	9	7	2
4	12	10	2

All three rows follow directly from Theorem 19's formula $\dim(R_k/I_k) = 3k - 2$; also confirmed by direct Gröbner-basis dimension computation in Sage at each k, as an independent check.

Proposition 21 ($Y_2 \not\subseteq Y_1$)

With Y_1, Y_2 as in the proof of Theorem 19: $Y_2 \not\subseteq Y_1$.

Proof of Proposition 21. Pick $p \in U_{k-1}$ (Lemma 8); then $\Phi(p, 0) = (p, 0) \in Y_2$ but $\overline{w}_{k-1}(p) \neq 0$ puts $(p, 0) \notin Y_1$. □

Proposition 22 (no component of Y_1 lies in Y_2)

No irreducible component Z of $V(I_{k-1})$ (each of dimension $3(k-1) - 2$, by Theorem 19 applied at $k-1$) has $Z \times \mathbb{A}^3 \subseteq Y_2$.

Proof of Proposition 22. Both $Z \times \mathbb{A}^3$ and Y_2 are irreducible: Y_2 by Lemma 15, and $Z \times \mathbb{A}^3$ because a product of irreducible varieties is irreducible (Z is irreducible by hypothesis, and $\mathbb{A}^3$ trivially is). Both have the **same** dimension $3k - 2$, by Theorem 19 and Lemma 18. Containment would therefore force equality. This is refuted: $Z \times \mathbb{A}^3$ is disjoint from $\pi^{-1}(U_{k-1})$, since $\overline{w}_{k-1}$ vanishes identically on $Z \subset V(I_{k-1})$; but Y_2 meets $\pi^{-1}(U_{k-1})$, by Lemma 18. □

Theorem 23 (component count)

For every $k \geq 2$: $V(I_k)$ has exactly $k - 1$ irreducible components, each of dimension $3k - 2$. The component indexed $j = 2, \ldots, k$ involves exactly the first j coordinate blocks and nothing later.

Proof of Theorem 23. Induction on k. **Base case $k = 2$**: the rank-≤ 1 locus of a generic 2×3 matrix is irreducible – the affine cone over the Segre embedding $\mathbb{P}^1 \times \mathbb{P}^2 \hookrightarrow \mathbb{P}^5$, equivalently the Motzkin-Taussky [3] irreducibility of the 2×2 commuting variety – so exactly 1 component, matching $k - 1 = 1$. **Step**: $V(I_k) = Y_1 \cup Y_2$ as in Theorem 19′s proof; by the inductive hypothesis Y_1 has $k - 2$ irreducible components, each of dimension $3k - 2$ (Theorem 19), none contained in another; Proposition 21 and Proposition 22 rule out containment between Y_2 and any of them or vice versa. A finite union of irreducible closed sets with no containments is irredundant, giving exactly $(k - 2) + 1 = k - 1$ components. □

This gives I_k's minimal primes and their count, but not yet that I_k is radical (so not yet that these are literally the components of a **primary** decomposition of I_k itself, as opposed to of $\sqrt{I_k}$) – that is closed in §7, after Cohen-Macaulayness is available.

Definition 24 (the components P_j)

For $k \geq 2$ and $j = 2, \ldots, k$, let Φ_j be Definition 14′s map Φ run at parameter j (with the base convention $\Phi_2(p, \lambda) := (p, \lambda p)$), and let

$$\Psi_j : \mathbb{A}^{3(j-1)} \times \mathbb{A}^1 \times \mathbb{A}^{3(k-j)} \to \mathbb{A}^{3k}, \quad (p, \lambda, q) \mapsto \big(\Phi_j(p, \lambda), q\big) \tag{32}$$

be Φ_j with the remaining blocks $\overline{v}_{j+1}, \ldots, \overline{v}_k$ carried along untouched. Since Ψ_j is a polynomial map it **is**, read backwards, a homomorphism of $\mathbb{F}$-algebras

$$\varphi_j : R_k \to \mathbb{F}[p, \lambda, q] = \mathbb{F}[t_1, \ldots, t_{3k-2}], \tag{33}$$

sending each variable $v_{i,c}$ to the corresponding coordinate polynomial of Ψ_j. Define

$$P_j := \ker \varphi_j \subset R_k. \tag{34}$$

Two consequences come for free. P_j is **prime**, being the kernel of a homomorphism into a domain – for every j including $j = 2$, with no case distinction and no appeal to Lemma 15, which is stated only for $k \geq 3$. And P_j does not depend on which $k \geq j$ it is viewed inside of: enlarging k appends free variables to source and target alike.

Why the kernel rather than the vanishing ideal. It is tempting to define P_j as the vanishing ideal of the j-th component of $V(I_k)$, and over $\overline{\mathbb{F}}$ the two agree – Proposition 26 identifies that component as the closure of Ψ_j's image. But over a **finite** field "vanishing ideal" is the wrong notion outright: $V(I_k)(\mathbb{F})$ is then a finite set of points, and $x^q - x$ vanishes on all of $\mathbb{A}^1(\mathbb{F}_q)$ without being the zero polynomial. The kernel is the right object over every field at once, is well defined with no convention attached, and is what makes the base-change statement in Remark 43 an identity rather than a coincidence. ♣

Example 25 (a suitable point of Y_2, concretely, at $k = 3$)

Rather than only the defining ideal (Example 28 below), a single explicit point: take $\overline{v}_1 = (1, 0, 0)$, $\overline{v}_2 = (0, 1, 0)$ (so $\overline{v}_1 \in U_2$, Lemma 8), giving $\overline{w}_2 = \overline{v}_1 \times \overline{v}_2 = (0, 0, 1)$ and $S(\overline{w}_2) = (0, 0, 1)$. Taking $\lambda = 1$ in Definition 14′s Φ, i.e. $\overline{v}_3 := S(\overline{w}_2) = (0, 0, 1)$, gives the point

$$(\overline{v}_1, \overline{v}_2, \overline{v}_3) = ((1, 0, 0), (0, 1, 0), (0, 0, 1)) \in Y_2 \subset \mathbb{A}^9. \tag{35}$$

Directly: $\overline{w}_3 = S(\overline{w}_2) \times \overline{v}_3 = (0, 0, 1) \times (0, 0, 1) = 0$ (a vector crossed with itself), confirming this point lies in $V(I_3)$, as any point of the form $\Phi(p, \lambda)$ must (Definition 14). The point lies on the **new** component P_3, not on the inherited P_2 – $\overline{v}_1$ and $\overline{v}_2$ are independent here, so the $k = 2$ bracket does not already vanish. It is **not** a rank-2 witness in Lemma 40′s sense (that witness uses $\lambda = 0$, not $\lambda = 1$) – it illustrates Y_2 itself, not the radicality argument's own witness points. ♢

Proposition 26 (rational parametrization of each component)

For $k \geq 2$ and $j = 2, \ldots, k$, the component of $V(I_k)$ indexed j (Theorem 23, over $\overline{\mathbb{F}}$ per Remark 43) is exactly the closure of the image of Definition 24′ map Ψ_j; equivalently, $P_j = \ker \varphi_j$ is its vanishing ideal. In particular that component is a **rational** variety. The parametrization is moreover **birational** onto the component: Ψ_j is injective over $U_{j-1} \times \mathbb{A}^1 \times \mathbb{A}^{3(k-j)}$ – where, at $j = 2$, the symbol U_1 is to be read as $\{\overline{v}_1 \neq 0\}$, since Lemma 8 defines U_j only for $j \geq 2$ and $\overline{w}_1$ does not exist. ♠

Proof of Proposition 26. At parameter $j = k$ the map Ψ_k is Φ itself, whose image closure is Y_2 (Definition 14), and Theorem 23's induction identifies Y_2 with the component indexed k. For $j < k$ the same construction run at parameter j gives that level's Y_2, and appending the free blocks $\overline{v}_{j+1}, \ldots, \overline{v}_k$ crosses it with $\mathbb{A}^{3(k-j)}$ – which is exactly Theorem 23's description of the inherited component indexed j. That the vanishing ideal of the closure of a polynomial map's image is the kernel of the corresponding algebra homomorphism is the standard identification, over $\overline{\mathbb{F}}$.

For birationality, Lemma 18's injectivity argument (over U_{j-1}, via Lemma 8 at level $j-1$) applies verbatim at parameter j, and appending the free blocks q unchanged does not affect injectivity. □

Remark 27 (a reducible variety is not itself "rational" – component-by-component is what Proposition 26 answers)

$Y_1 := V(I_{k-1}) \times \mathbb{A}^3$ (Definition 14) is **not** irreducible for $k \geq 4$ – it is a union of $k-2$ of the components above (Theorem 23's inductive step) – so a single rational parametrization **of Y_1 as one object** is not meaningful in the usual sense (rationality is a property of irreducible varieties). What **is** meaningful, and what Proposition 26 supplies, is a rational parametrization of each of Y_1's irreducible pieces individually – exactly the components inherited from $V(I_{k-1})$.

◇

Example 28 (the components, worked out at $k = 3$)

Uniformly by k, since the components' **count** and **block pattern** are already known (Theorem 23, Definition 24) and only their explicit generators change:

- **At $k = 2$**: I_2 is already prime (Theorem 23's base case), so $V(I_2)$ has exactly $k - 1 = 1$ component – itself.
- **At $k = 3$**: exactly $k - 1 = 2$ components, both computed directly (Gröbner bases, not inferred from the pattern alone), matching $I_3 = P_2 \cap P_3$:
  - P_2, the "Y_1" piece (block pattern $\{1, 2\}$, matching that the length-2 sub-bracket already vanishes with $\overline{v}_3$ free) – generated by exactly I_2's three generators from the first example, now inside R_3 with $\overline{v}_3$ simply not appearing:
  $$v_{1,2}v_{2,3} - v_{1,3}v_{2,2}, \quad v_{1,3}v_{2,1} - v_{1,1}v_{2,3}, \quad v_{1,1}v_{2,2} - v_{1,2}v_{2,1}. \tag{36}$$
  - P_3, the "Y_2" piece (block pattern $\{1, 2, 3\}$) – exactly the ideal already computed in the Y_2-defining-ideal example above:
  $$v_{2,1}v_{3,1} + v_{2,2}v_{3,2} - v_{2,3}v_{3,3}, \quad v_{1,1}v_{3,1} + v_{1,2}v_{3,2} - v_{1,3}v_{3,3}, \tag{37}$$
  $$v_{1,2}v_{2,1}v_{3,2} - v_{1,1}v_{2,2}v_{3,2} - v_{1,3}v_{2,1}v_{3,3} + v_{1,1}v_{2,3}v_{3,3}. \tag{38}$$
- **At $k = 4$**: exactly $k - 1 = 3$ components, block patterns $\{1, 2\}$, $\{1, 2, 3\}$, $\{1, 2, 3, 4\}$:
  - The first two, P_2 and P_3, are inherited **verbatim** from $k = 3$ above, with $\overline{v}_4$ simply left free – not recomputed, per Definition 24′ own "does not depend on which k" clause.
  - The third, P_4, is the new "Y_2" stratum – exactly the $k = 4$ row of the earlier Y_2-defining-ideal example (generators omitted here to avoid repeating that display; same growth pattern, quartic with up to 8 terms per generator).

◇

6. Cohen-Macaulayness

Proposition 29 (Eagon-Northcott / Hilbert-Burch, applied)

Let M be a 2×3 matrix with entries in a polynomial ring R over a field (arbitrary entries, not necessarily independent indeterminates), and J the ideal of its 2×2 minors. If J has the **expected** codimension $3 - 2 + 1 = 2$, then J is a perfect ideal of grade 2 – $\mathrm{pd}(R/J) = 2$, hence R/J is Cohen-Macaulay.

♠

Proof of Proposition 29. Bruns and Vetter [1], Theorem (2.7), p. 13, gives exactly this for an **arbitrary** matrix, as needed since one row of our 2×3 matrix $(S(\overline{w}_{k-1}))$ is a fixed polynomial vector, not a row of independent indeterminates. (The original construction, Eagon and Northcott [2], is superseded here by Bruns-Vetter's more general treatment – confirmed closed-access via Unpaywall, not used directly.) □

Remark 30 (why the generator count above wasn't proven this way)

It is tempting to expect Proposition 29 to also hand over the generator count for free via Bruns-Vetter's Theorem (16.36), p. 217 ("The Theorem of Hilbert-Burch") – but that theorem, as actually stated, gives only $I = a \cdot I_m(f)$ for a nonzerodivisor a, not a generator count on its own: at expected codimension, an **arbitrary** 2×3 matrix's minors can be perfect and Cohen-Macaulay while one minor vanishes identically or is a scalar combination of the others (concretely: $\begin{pmatrix} x & y & 0 \\ 0 & 0 & 1 \end{pmatrix}$ over $\mathbb{F}[x, y]$ has minors $0, x, y$, height 2 as expected, $J = (x, y)$ perfect and Cohen-Macaulay with only **2** generators). This is exactly why Theorem 12 was proven directly from linear independence instead of cited from this section's machinery. ◇

Theorem 31 (R_k/I_k is Cohen-Macaulay)

For every $k \geq 2$: R_k/I_k is Cohen-Macaulay.

Proof of Theorem 31. By Theorem 19, I_k has codimension 2, the expected value for the 2×3 matrix of Theorem 6 presenting it. Proposition 29 applies directly. (This uses Theorem 19 only – not Theorem 12's generator count, not Theorem 23's component count – genuinely independent of both.) □

7. Radicality

Cohen-Macaulayness and the explicit component structure of §5 combine, via Serre's criterion, to give radicality – which then closes both the generator-count (§4) / Cohen-Macaulay (§6) story (making I_k itself, not just $\sqrt{I_k}$, the object with those properties) and upgrades §5's component count to a genuine primary decomposition.

Lemma 32 (reduction via Serre)

Recall Serre's condition S_n: a Noetherian ring A satisfies S_n if $\operatorname{depth}(A_P) \geq \min(n, \dim A_P)$ for every prime $P \subset A$ (Cohen-Macaulay means $\operatorname{depth}(A_P) = \dim A_P$ at **every** prime – the extremal case, $n = \infty$ in effect). Cohen-Macaulay rings satisfy S_n for every n: $\operatorname{depth}(A_P) = \dim A_P \geq \min(n, \dim A_P)$ trivially, whatever n is – this needs nothing beyond the two definitions just given, standard for Serre's conditions in general (see e.g. Bruns and Vetter [1] or Eisenbud's *Commutative Algebra with a View Toward Algebraic Geometry*). In particular S_1. Serre's criterion for reducedness – a Noetherian ring is reduced if and only if it satisfies R_0 and S_1 [12] – then says that, given Theorem 31, radicality of I_k is equivalent to R_0 alone: regularity of R_k/I_k at each of its minimal primes.

The four lemmas that follow are the standard Jacobian criterion, split into its two genuinely separate ingredients – one linear-algebraic (Lemma 33), one dimension-theoretic (Lemma 34) – before either is used. The order is deliberate: the rank bound and the regularity criterion each draw on **both** ingredients and on neither each other, which a single pair of mutually-citing lemmas (as in an earlier draft of this section) obscures.

Lemma 33 (the cotangent identity)

Let $S = \mathbb{F}[x_1, \ldots, x_n]$, $I = (f_1, \ldots, f_m)$, $A = S/I$, and $x \in V(I)$ an $\mathbb{F}$-rational point, with $\mathfrak{n} \subset S$ the maximal ideal at x and $\mathfrak{m} = \mathfrak{n}/I \subset A$ its image. Then

$$\dim_{\mathbb{F}} \mathfrak{m}/\mathfrak{m}^2 = n - \operatorname{rank} \operatorname{Jac}(x). \tag{39}$$

Proof of Lemma 33. Linear algebra only; no hypothesis on A is needed. Sending g to its gradient at x identifies the cotangent space $\mathfrak{n}/\mathfrak{n}^2$ with $\mathbb{F}^n$. Under that identification the image of I is the span of the linear parts at x of $f_1, \ldots, f_m$: a general element of I is $\sum_i g_i f_i$, whose linear part at x is $\sum_i g_i(x)$ times that of f_i, since every f_i vanishes at x. That span is the row space of $\operatorname{Jac}(x)$, of dimension rank $\operatorname{Jac}(x)$. As $\mathfrak{m}/\mathfrak{m}^2 \cong (\mathfrak{n}/\mathfrak{n}^2)/\operatorname{im}(I)$, the claim follows. □

Lemma 34 (the local dimension at a rational point)

Let $S = \mathbb{F}[x_1, \ldots, x_n]$, $I = (f_1, \ldots, f_m)$, $A = S/I$ equidimensional of dimension $n - c$, and $x \in V(I)$ an $\mathbb{F}$-rational point with maximal ideal $\mathfrak{m} \subset A$ and Jacobian $\operatorname{Jac}(x) \in \mathbb{F}^{m \times n}$. Then $\dim A_{\mathfrak{m}} = n - c$.

Proof of Lemma 34. $\dim A_{\mathfrak{m}} = \max\{\dim (A/P)_{\mathfrak{m}} : P \in \operatorname{Min}(A), P \subseteq \mathfrak{m}\}$, and each $A/P = S/\tilde{P}$ is an affine domain over a field, for which $\dim \left(S/\tilde{P}\right)_{\mathfrak{M}} = \dim S/\tilde{P}$ at **every** maximal ideal $\mathfrak{M}$ – affine domains over a field are catenary and equicodimensional (Eisenbud, *Commutative Algebra*, Cor. 13.4; equivalently the Stacks Project, tag 00OS [4] – a dimension formula for finitely generated field extensions, holding over an arbitrary field, not needing a perfect-field hypothesis by accident). By equidimensionality every such $\dim S/\tilde{P}$ equals $n - c$. □

Lemma 35 (Jacobian criterion: the rank bound)

Hypotheses as in Lemma 34. Then $\operatorname{rank} \operatorname{Jac}(x) \leq c$.

Proof of Lemma 35. $\dim_{\mathbb{F}} \mathfrak{m}/\mathfrak{m}^2 \geq \dim A_{\mathfrak{m}}$ in any Noetherian local ring. By Lemma 33 the left-hand side is $n - \operatorname{rank} \operatorname{Jac}(x)$, and by Lemma 34 the right-hand side is $n - c$; rearranging gives $\operatorname{rank} \operatorname{Jac}(x) \leq c$. □

Lemma 36 (Jacobian criterion: regularity at rank $= c$)

Hypotheses as in Lemma 34. If $\operatorname{rank} \operatorname{Jac}(x) = c$, then $A_{\mathfrak{m}}$ is regular of dimension $n - c$; in particular x lies on a **unique** component of $V(I)$, whose local ring there is a field.

Proof of Lemma 36. If $\operatorname{rank} \operatorname{Jac}(x) = c$ then $\dim_{\mathbb{F}} \mathfrak{m}/\mathfrak{m}^2 = n - c$ by Lemma 33, while $\dim A_{\mathfrak{m}} = n - c$ by Lemma 34; equality of the two is the definition of regularity. A regular local ring is a domain, so $\mathfrak{m}$ contains a unique minimal prime of A, and the local ring at that prime is a further localization of a regular local ring at its own maximal ideal in dimension 0, hence a field. □

Lemma 37 (the Jacobian's own recursion)

Write J_k for the $3 \times 3k$ Jacobian of $\overline{w}_k$, $[u]_\times$ for the skew matrix with $[u]_\times v = u \times v$. Then $J_2 = \left[-[\overline{v}_2]_\times \;\middle|\; [\overline{v}_1]_\times\right]$, and for $k \geq 3$,

$$J_k = \left[-[\overline{v}_k]_\times S J_{k-1} \;\middle|\; [S(\overline{w}_{k-1})]_\times\right]. \tag{40}$$

Proof of Lemma 37. For the second block, $\overline{w}_k = S(\overline{w}_{k-1}) \times \overline{v}_k = [S(\overline{w}_{k-1})]_\times \overline{v}_k$ is linear in $\overline{v}_k$ with coefficient matrix free of $\overline{v}_k$. For the first, differentiating $\overline{w}_k = S(\overline{w}_{k-1}) \times \overline{v}_k$ with respect to any variable t among $\overline{v}_1, \ldots, \overline{v}_{k-1}$ gives $\partial_t \overline{w}_k = (S\partial_t \overline{w}_{k-1}) \times \overline{v}_k = -[\overline{v}_k]_\times S\partial_t \overline{w}_{k-1}$, using $a \times b = -[b]_\times a$; assembling columns gives the stated product. □

Example 38 (J_3, explicitly)

With variables ordered $v_{1,1}, v_{1,2}, v_{1,3}, v_{2,1}, v_{2,2}, v_{2,3}, v_{3,1}, v_{3,2}, v_{3,3}$, the 3×9 Jacobian of $\overline{w}_3$ (the three generators of I_3 from the earlier example), column-blocked to match Lemma 37's formula $J_3 = \left[-[\overline{v}_3]_\times S J_2 \;\middle|\; [S(\overline{w}_2)]_\times\right]$ – shown here as the two blocks separately, since side by side the full 3×9 matrix does not fit the page width. First, $-[\overline{v}_3]_\times S J_2$ (3×6):

$$\begin{pmatrix} -v_{2,2}v_{3,2} + v_{2,3}v_{3,3} & v_{2,1}v_{3,2} & -v_{2,1}v_{3,3} & v_{1,2}v_{3,2} - v_{1,3}v_{3,3} & -v_{1,1}v_{3,2} & v_{1,1}v_{3,3} \\ v_{2,2}v_{3,1} & -v_{2,1}v_{3,1} + v_{2,3}v_{3,3} & -v_{2,2}v_{3,3} & -v_{1,2}v_{3,1} & v_{1,1}v_{3,1} - v_{1,3}v_{3,3} & v_{1,2}v_{3,3} \\ -v_{2,3}v_{3,1} & -v_{2,3}v_{3,2} & v_{2,1}v_{3,1} + v_{2,2}v_{3,2} & v_{1,3}v_{3,1} & v_{1,3}v_{3,2} & -v_{1,1}v_{3,1} - v_{1,2}v_{3,2} \end{pmatrix} \tag{41}$$

and then $[S(\overline{w}_2)]_\times$ (3×3, the skew matrix of $S(\overline{w}_2)$'s three coordinates from the earlier matrix example):

$$\begin{pmatrix} 0 & v_{1,2}v_{2,1} - v_{1,1}v_{2,2} & -v_{1,3}v_{2,1} + v_{1,1}v_{2,3} \\ -v_{1,2}v_{2,1} + v_{1,1}v_{2,2} & 0 & -v_{1,3}v_{2,2} + v_{1,2}v_{2,3} \\ v_{1,3}v_{2,1} - v_{1,1}v_{2,3} & v_{1,3}v_{2,2} - v_{1,2}v_{2,3} & 0 \end{pmatrix} \tag{42}$$

Both blocks computed directly, not merely asserted from the recursive formula.

Remark 39 (a false shortcut, caught and corrected)

A naive reading of Lemma 37 suggests appending $\overline{v}_k = 0$ preserves rank on an inherited component – this is false: the old-variable block is left-multiplied by $-[\overline{v}_k]_\times S$, not merely padded with zero rows. At the first inherited case ($k = 2 \to 3$), appending $\overline{v}_3 = 0$ actually drops the rank to 0. The correct witness (Lemma 41 below) instead picks $\overline{v}_k = e_i$ for a standard basis vector **not** lying in $S \cdot \operatorname{im} J_{k-1}(q)$ – a genuine choice, not a formality.

Lemma 40 (rank-2 witness on the new component)

For every $k \geq 3$, there is a point $x = (p, 0)$, p the Lemma 8 non-degeneracy witness at level $k-1$, with $x \in V(I_k)$ and rank $J_{k(x)} = 2$. At $k = 2$ the same conclusion holds with $x = (p, 0)$ for any $p \neq 0$.

Proof of Lemma 40. **The case $k = 2$ separately**, because the statement above cannot be read at $k = 2$: it would refer to a non-degeneracy witness "at level 1", and neither $\overline{w}_1$ nor Lemma 8 exists there — Definition 3 starts the recursion at $j = 2$. Nor does Lemma 37's $k \geq 3$ formula apply. Argue directly instead: $J_2 = \left[-[\overline{v}_2]_\times \;\middle|\; [\overline{v}_1]_\times\right]$ carries no S at all, so for any $p \neq 0$, $J_2(p, 0) = [0 \mid [p]_\times]$, which has rank 2 by the kernel computation below. And $(p, 0) \in V(I_2)$ since $\overline{w}_2(p, 0) = p \times 0 = 0$.

For $k \geq 3$. By Lemma 37, $J_{k(p,0)} = \left[0 \;\middle|\; [S(\overline{w}_{k-1}(p))]_\times\right]$ since $-[0]_\times S J_{k-1}(p) = 0$; and $S(\overline{w}_{k-1}(p)) \neq 0$ since $\overline{w}_{k-1}(p) \neq 0$ (Lemma 8) and S is invertible. For $u \neq 0$, $\ker [u]_\times = \{v : u \times v = 0\} = \mathbb{F}u$ (a cross product vanishes exactly on linearly dependent pairs), so $[S(\overline{w}_{k-1}(p))]_\times$ has rank 2, giving rank $J_{k(p,0)} = 2$. □

Lemma 41 (rank-2 witness on an inherited component)

For every $k \geq 3$ and every component Z of $V(I_{k-1})$ with a rank-2 witness $q \in Z$ (Lemma 40, or this lemma applied at $k-1$), there is $x = (q, e_i) \in Z \times \mathbb{A}^3 \subset V(I_k)$ with rank $J_{k(x)} = 2$.

Proof of Lemma 41. Let $W := S \cdot \operatorname{im} J_{k-1}(q)$, a 2-dimensional subspace of $\mathbb{F}^3$ since S is invertible and rank $J_{k-1}(q) = 2$. Since e_1, e_2, e_3 span $\mathbb{F}^3$ and $W \neq \mathbb{F}^3$, some $e_i \notin W$; fix such an i. Since $q \in V(I_{k-1})$, $\overline{w}_{k-1}(q) = 0$, so Lemma 37 gives $J_{k(q,e_i)} = \left[-[e_i]_\times S J_{k-1}(q) \;\middle|\; 0\right]$, of rank $\dim\left([e_i]_\times W\right)$ since $S J_{k-1}(q)$ has image exactly W. As $\ker [e_i]_\times = \mathbb{F}e_i$ and $W \cap \mathbb{F}e_i = 0$ (else $e_i \in W$), $[e_i]_\times$ is injective on W, so $\dim\left([e_i]_\times W\right) = 2$. □

Theorem 42 (I_k is radical)

For every $k \geq 2$: I_k is radical.

Proof of Theorem 42. By induction on k (mirroring Theorem 23's own induction), Lemma 40 and Lemma 41 exhibit a rank-2-Jacobian point on **every** one of the $k-1$ components of Theorem 23. Lemma 36 then gives R_0 at every one (and, by its "unique component" clause, confirms each witness lies only on its intended component – an independent check on Theorem 23's count, not merely assumed); Lemma 32 gives S_1, using Theorem 31's Cohen-Macaulayness. Serre's criterion [12] gives radicality. □

Verified independently in Sage, for the signed encoding, at every $k = 2, \ldots, 6$: the Jacobian recursion (Lemma 37) checked as a polynomial identity, not just at points, and every witness confirmed to lie on $V(I_k)$ with rank $J_k = 2$ (*code/referee_radicality_D_check.sage*). The sharper claim — that each witness lies on **exactly** its intended component — is confirmed by an independently-computed primary decomposition at every $k = 3, 4, 5, 6$, which returns $k-1$ components in each case, in agreement with Theorem 23; the $k = 6$ decomposition takes some hours.

8. The Primary Decomposition Theorem

Every ingredient is now in hand for the fully general statement – component count (§5) and radicality (§7) together, neither sufficient alone. One convention has to be made explicit first, because it is the hinge on which the statement's claimed generality turns.

Remark 43 (the geometric arguments run over $\overline{\mathbb{F}}$, and the conclusion descends)

Sections 5-7 argue geometrically: they speak of $V(I_k)$, of its irreducible components, and of the closure of the image of a parametrization, and at the decisive step they identify $\mathrm{Min}(I_k)$ with the vanishing ideals of those components. That identification is a Nullstellensatz statement and is **false over a general field** – over $\mathbb{F}_3$, for instance, $V(I_k)(\mathbb{F}_3)$ is a finite set of points and carries no such information at all. So throughout §§5-7, and in the theorem below, $V(\cdot)$ means the vanishing locus in $\mathbb{A}^{3k}\left(\overline{\mathbb{F}}\right)$ over a fixed algebraic closure, and "component" means an irreducible component there. This is the usual convention and is what the classical literature on the commuting variety assumes as well (Artin-Hochster is stated over an algebraically closed field in [7]).

The conclusion nonetheless descends to $\mathbb{F}$ itself, so the theorem may be read over any field of characteristic $\neq 2$, as stated. The reason is that each P_j is not merely prime but **absolutely** prime. By Definition 24, P_j **is** the kernel of an $\mathbb{F}$-algebra homomorphism $\varphi_j : R_k \to \mathbb{F}[t_1, \ldots, t_{3k-2}]$ into a polynomial ring, so R_k/P_j embeds in that polynomial ring. Since $\overline{\mathbb{F}}$ is flat over $\mathbb{F}$, tensoring preserves the embedding:

$$\overline{\mathbb{F}} \otimes_{\mathbb{F}} \left(R_k/P_j\right) \hookrightarrow \overline{\mathbb{F}}[t_1, \ldots, t_m], \tag{43}$$

whose target is a domain, so the extension of P_j to $\overline{\mathbb{F}} \otimes_{\mathbb{F}} R_k$ is again prime. Formation of a finite intersection of ideals likewise commutes with the flat base change $\mathbb{F} \to \overline{\mathbb{F}}$. Hence the displayed decomposition, established over $\overline{\mathbb{F}}$, holds already over $\mathbb{F}$, with the same $k-1$ primes.

Theorem 44 (Primary Decomposition Theorem)

For every $k \geq 2$, over any field of characteristic $\neq 2$: I_k has exactly $k-1$ irreducible, radical, primary components, namely $P_2, \ldots, P_k$ of Definition 24, and

$$I_k = P_2 \cap \cdots \cap P_k \tag{44}$$

is an irredundant primary decomposition into $k-1$ primes.

Proof of Theorem 44. Spelled out in full, combining the two independent halves proven above rather than left as a one-line conjunction. By Theorem 42, I_k is radical, i.e. $I_k = \sqrt{I_k}$. A standard fact about radical ideals in a Noetherian ring: a radical ideal equals the intersection of its own minimal primes, $I_k = \cap_{P \in \mathrm{Min}(I_k)} P$ – since $\sqrt{I_k}$ equals the intersection of **all** primes containing I_k (Krull's theorem, not the definition of the radical, which is the set of elements with a power in I_k), and the minimal ones already suffice: every non-minimal prime containing I_k contains some minimal one, hence is redundant in the intersection. Theorem 23 identifies $\mathrm{Min}(I_k)$ explicitly – over $\overline{\mathbb{F}}$, per Remark 43 – as exactly $k-1$ primes, one per block pattern $j = 2, \ldots, k$, named $P_2, \ldots, P_k$ by Definition 24. Substituting gives the displayed equation. Each P_j, being prime, is trivially its own primary-decomposition component (P_j-primary with associated prime P_j itself) – no separate argument for "primary" beyond "prime" is needed. For primality itself, nothing is left to prove: Definition 24 **defines** P_j as $\ker \varphi_j$, and the kernel of a homomorphism into a domain is prime. This covers $j = 2$ along with the rest, at no extra cost. (Lemma 15 gives the same conclusion for $j \geq 3$ but not for $j = 2$, being stated for $k \geq 3$; P_2 is I_2 itself, the ideal of maximal minors of a generic 2×3 matrix, classically prime. Neither route is needed now.) Extending P_j from R_j to R_k by adjoining the free blocks $\overline{v}_{j+1}, \ldots, \overline{v}_k$ preserves primality, since R_k/P_jR_k is a polynomial ring over the domain R_j/P_j. Irredundancy – no P_j can be omitted, i.e. no P_j contains $\cap_{i \neq j} P_i$ – is exactly Theorem 23's own no-containment conclusion (Proposition 21 and Proposition 22, applied at every level of that theorem's induction), not a fact needing any further proof here. □

Remark 45 (summary)

Theorem 19 (codimension 2), Theorem 12 (3 minimal generators), Theorem 31 (Cohen-Macaulay), and Theorem 42 (radical) together say exactly what a single bundled "I_k has codimension 2, is radical, has exactly 3 minimal generators, and is Cohen-Macaulay" statement would – now as four independently-proven results, each established at the point in this article where its actual dependencies (and no others) were available, rather than one claim proven in parts across several sections.

9. Summary

For every $k \geq 2$, over any field $\mathbb{F}$ of characteristic $\neq 2$:

- I_k has codimension 2 (Theorem 19), presented as the ideal of 2×2 minors of an explicit 2×3 matrix (Theorem 6) – a genuine determinantal ideal, not merely determinantal-looking.
- I_k has exactly 3 minimal generators (Theorem 12), proven directly from linear independence, independent of the determinantal presentation.
- R_k/I_k is Cohen-Macaulay (Theorem 31), from the determinantal presentation and classical Eagon-Northcott/ Hilbert-Burch theory (Proposition 29).
- I_k is radical (Theorem 42), via Serre's criterion: Cohen-Macaulayness gives S_1 (Lemma 32), and an explicit rank-2 Jacobian witness on every component gives R_0 (Lemma 40, Lemma 41).
- $V(I_k)$ has exactly $k-1$ irreducible components (Theorem 23), each rational – in fact birational to an explicit affine parameter space (Proposition 26). Radicality and this component count together – rationality plays no part in the assembly – give the irredundant primary decomposition of the Primary Decomposition Theorem (Theorem 44): $I_k = P_2 \cap \cdots \cap P_k$.

10. Methods

This article was produced by a workflow that is not the usual one. Since the workflow bears on how some of the claims above should be read, it is set out here rather than left to be inferred.

Computer algebra, in two roles that should not be confused. SageMath and Macaulay2 were used throughout, for two quite different purposes. The first is **conjecture**: the component count of Theorem 23, its block-involvement pattern, and the shape of the primary decomposition were found by computing primary decompositions of I_k at small k and reading off the pattern, well before any of it was proved. The second is **verification**: once proved, the general statements were checked against independent computation — the determinantal structure of Theorem 6 by Gröbner-basis equality in both directions and, separately, against ordinary 2×2 matrix multiplication, the two routes sharing no code; the Jacobian recursion of Lemma 37 as a polynomial identity rather than at sampled points; the witness points of Lemma 40 and Lemma 41 confirmed to lie on $V(I_k)$ with the stated Jacobian rank, and on exactly the intended component; and the decomposition itself recomputed at $k = 3, 4, 5, 6$.

Because rank, dimension and radicality are **not** characteristic-independent — a matrix of rank 3 over $\mathbb{Q}$ may drop rank modulo p — computations over $\mathbb{Q}$ license nothing about $\mathbb{F}_p$, and the four main results were therefore re-run over $\mathbb{F}_2$, $\mathbb{F}_3$, $\mathbb{F}_4$, $\mathbb{F}_5$, $\mathbb{F}_7$ and $\mathbb{F}_9$ as well. No deviation appeared. The scripts are named at the point of use and are kept, with their output, alongside the source.

Nothing in this article is conjectural, or checked only at small k: every statement is proven for **every** $k \geq 2$, over an **arbitrary** field of the stated characteristic. The computational thread corroborates the general proofs; nowhere does it substitute for one.

Source of truth, and a partial formalization. The manuscript is written in Typst, and that source is canonical: the LaTeX is generated from it mechanically, so the two cannot drift apart. The same source supplied the statements for a partial formalization in Lean 4 against *mathlib*. Machine-checked, with no proof gaps and depending on no axioms beyond the three that every *mathlib* development uses: the basic identity (Theorem 1), in the strengthened form that drops the traceless hypothesis; the minimal-generator theorem (Theorem 12) for the actual $\overline{w}_k$, together with the whole chain beneath it — the general lemma that independent same-degree generators are minimal (Lemma 9), the homogeneity of $\overline{w}_k$, the freshness of each new variable block, and the linear independence itself (Lemma 10); the Jacobian recursion (Lemma 37); and the smaller computational claims of §3 and §5, including the identity of Remark 2.

Not formalized: everything resting on determinantal-ideal theory or on dimension theory — the codimension (Theorem 19), Cohen-Macaulayness (Theorem 31), radicality (Theorem 42), the component count (Theorem 23) and the Primary Decomposition Theorem (Theorem 44) — for which the necessary libraries do not presently exist. The formalization is therefore a **partial** guarantee, and is described as one: it should not be read as covering the article's main results.

Adversarial review. The manuscript was put through five rounds of deliberately adversarial review, each instructed to find errors rather than to approve, and each required to verify by computation rather than by reading. This was not a formality. The rounds found: a genuine gap in the proof of Lemma 10, where non-degeneracy was doing work that only linear independence can do; explicit displays that had not been recomputed after a change of encoding, and so still showed the superseded values; a false statement about the state of the literature, contradicted

by the very sources cited for it; a citation to the wrong tag of a reference work, made under an explicit assurance that it had been verified; a stretch of prose that described the formulas printed beside it incorrectly; and the gap in generality that Remark 43 now closes. All are corrected above. The mention is not decorative: a reader is entitled to know what kind of error an adversarial pass actually catches, and to calibrate accordingly.

What this workflow is, and is not, good for. Its strength is coverage of the mechanical: a recomputation notices a stale display, an inconsistent index or an unverified citation reliably, and does so at a scale and patience that re-reading does not sustain. Its weakness is that none of that establishes that the argument is **right**; every substantive correction above was found by an adversarial reading, not by a passing test. The machine checks and the Lean formalization narrow the space in which an error can hide; they do not empty it, and the responsibility statement below is meant literally.

11. Acknowledgements

The author first learned of the ring of commuting matrices some thirty years ago from Professor Jan-Erik Roos, in a course he held on the then fledgling field of computer algebra. It is a pleasant debt to record here, given how much of what follows was found with a computer algebra system open beside the page.

The computations were carried out in SageMath and Macaulay2, and the formalization in Lean 4 with *mathlib*; this article would have a different and smaller content without them. The nature and extent of the assistance from a large language model is set out in §10 and in the disclosure below.

12. Disclosure of AI assistance

This manuscript was produced with substantial assistance from a large language model, Claude (Anthropic), and the extent of it is stated here rather than left to be inferred. The itemised account is §10: which results were machine-checked and which were not, what the computer algebra established and what it merely corroborated, and what the adversarial review found. Claude is not a co-author, and is not credited as one.

Every computational claim has been checked along a route independent of the statement being tested, and the reference list has been checked against the sources cited — with the one exception that where a source is a printed book outside this project's open-access policy, an independently verified equivalent is cited alongside it. The author has verified the results to the best of his ability, made the final edits, and assumes full responsibility for any errors, mistakes or gaps that remain.

13. References

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[11] A. Youcis, answer to "What can we say about the image of a regular map?", Mathematics Stack Exchange, question 731178, answer 732415, 2014. States both the point-set-topology fact that the continuous image of an irreducible space is irreducible, and the $(x, y) \mapsto (x, xy)$ non-closed-image example cited above. CC BY-SA 3.0. Open access at math.stackexchange.com/questions/731178.

[12] *The Stacks Project*, tag **031R** = Lemma 10.157.3, Serre's criterion for reducedness: "Let R be a Noetherian ring. The following are equivalent: (1) R is reduced, and (2) R has properties (R_0) and (S_1)." Open access at stacks.math.columbia.edu/tag/031R. Note for anyone re-checking: the enclosing **section** 10.157 is tag 031O and is titled "Serre's criterion for *normality*"; its own Lemma 10.157.4 (tag 031S) is the normality criterion $R_1 + S_2$, a different statement. Fetching 031O displays the whole section, this lemma included, so it is easy to confirm the quoted text on a page whose tag is nevertheless the wrong one – which is exactly the error this entry carried until 2026-08-31.

[13] T. Tambour, *The Number of Solutions of Some Equations in Finite Groups and a New Proof of Itô's Theorem*, Communications in Algebra **28** (2000), no. 11, 5353–5361. doi:10.1080/00927870008827160. Earlier version: *On the Number of Solutions of Some Equations in Finite Groups*, Stockholm University Research Reports in Mathematics No. 15 (1998), ISSN 1401-5617.